

Logical Toolkit for Applied Ontology

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Propositional Logic

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First-Order Logic

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Description Logic

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SHIQ

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Satisfiability

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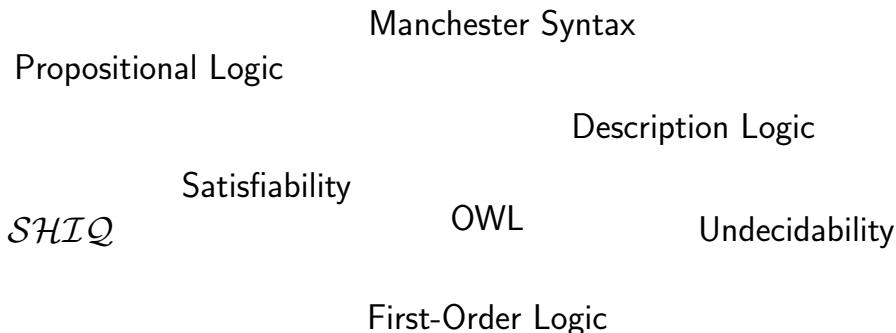
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Why Use Logic in Applied Ontology?

- **Precise formalisation:** Logic provides a rigorous tool to define concepts, relationships, and constraints.
- **Automated reasoning:** Enables automated inference and consistency checking.
- **Built-in “explainability”:** Every conclusion can be traced back to axioms and rules.

Outline

- 1 First-Order Logic (FOL)
- 2 Description Logics (DLs)
- 3 OWL Ontologies in the Manchester Syntax

First-Order Logic (FOL)

$$\forall s \left((Student(s) \wedge AttendsISAO(s)) \rightarrow \exists \ell (Logic(\ell) \wedge Understands(s, \ell)) \right)$$

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$$\begin{aligned} &\exists t \left(Teacher(t) \wedge AttendsISAO(t) \wedge \right. \\ &\quad \left. \forall s \forall \ell ((Teaches(t, s) \wedge Understands(s, \ell)) \rightarrow Understands(t, \ell)) \right) \end{aligned}$$

Syntax of First-Order Logic

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- **Constants:** $a, b, c, \dots$ (refer to specific objects)
- **Variables:** $x, y, z, \dots$ (range over the domain)

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$$\varphi ::= P(t_1, \dots, t_n) \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \rightarrow \varphi \mid \forall x \varphi \mid \exists x \varphi \mid t_1 = t_2$$

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- **Terms:** Constants or Variables
- **Constants:** $a, b, c, \dots$ (refer to specific objects)
- **Variables:** $x, y, z, \dots$ (range over the domain)
- **Predicates:** $P, Q, R, \dots$ (properties and relations)
- **Connectives:** $\neg$ (not), $\wedge$ (and), $\vee$ (or), $\rightarrow$ (if...then),
- **Quantifiers:** $\forall x$ (for all x), $\exists x$ (there exists x)

Examples:

- “Every human is mortal” $\Rightarrow \forall x(\text{Human}(x) \rightarrow \text{Mortal}(x))$
- “Socrates is a human” $\Rightarrow \text{Human}(\text{Socrates})$
- “Some cats like Alice” $\Rightarrow \exists x(\text{Cat}(x) \wedge \text{Likes}(x, \text{Alice}))$
- “If it is raining, the ground is wet” $\Rightarrow \text{Raining} \rightarrow \text{Wet}(\text{Ground})$

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Exercise: “Every student has taken at least one course, and every course has at least one teacher.”

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Exercise: “Every student has taken at least one course, and every course has at least one teacher.”

$$\forall x(\text{Student}(x) \rightarrow \exists c(\text{Course}(c) \wedge \text{Takes}(x, c))) \wedge$$
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Model Theory connects syntax (formulas) with semantics (meaning);
Truth is relative to an interpretation.

Semantics of First-Order Logic

- An **interpretation** $\mathcal{I} = \langle \Delta^{\mathcal{I}}, \cdot^{\mathcal{I}} \rangle$ consists of:
 - $\Delta^{\mathcal{I}}$: a non-empty domain of objects
 - $\cdot^{\mathcal{I}}$: an interpretation function mapping
 - constants c to elements $c^{\mathcal{I}} \in \Delta^{\mathcal{I}}$
 - n -ary predicates P to relations $P^{\mathcal{I}} \subseteq (\Delta^{\mathcal{I}})^n$
- A **valuation** assigns domain elements to variables: $x \mapsto d \in \Delta^{\mathcal{I}}$

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Mary

Math

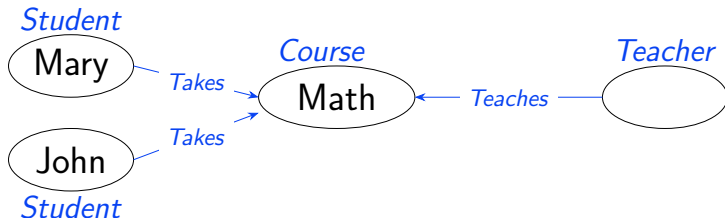
John

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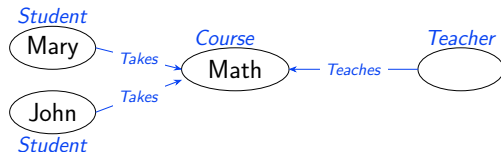
First-Order Logic

Satisfaction in First-Order Logic Interpretations

Satisfaction $\mathcal{I}, v \models \varphi$ is defined inductively:

- $\mathcal{I}, v \models P(t_1, \dots, t_n)$ iff $(t_1^{\mathcal{I}}, \dots, t_n^{\mathcal{I}}) \in P^{\mathcal{I}}$
- $\mathcal{I}, v \models \neg\varphi$ iff $\mathcal{I}, v \not\models \varphi$
- $\mathcal{I}, v \models \varphi \wedge \psi$ iff $\mathcal{I}, v \models \varphi$ and $\mathcal{I}, v \models \psi$
- $\mathcal{I}, v \models \varphi \vee \psi, \varphi \rightarrow \psi, \forall x \varphi, \exists x \varphi$ similarly

Model: An interpretation $\mathcal{I}$ is a model of φ if $\mathcal{I} \models \varphi$



$$\begin{aligned}\varphi := & \forall x (\text{Student}(x) \rightarrow \\ & \exists c (\text{Course}(c) \wedge \text{Takes}(x, c))) \wedge \\ & \forall c (\text{Course}(c) \rightarrow \\ & \exists t (\text{Teacher}(t) \wedge \text{Teaches}(t, c)))\end{aligned}$$

Back to the Introductory Lecture

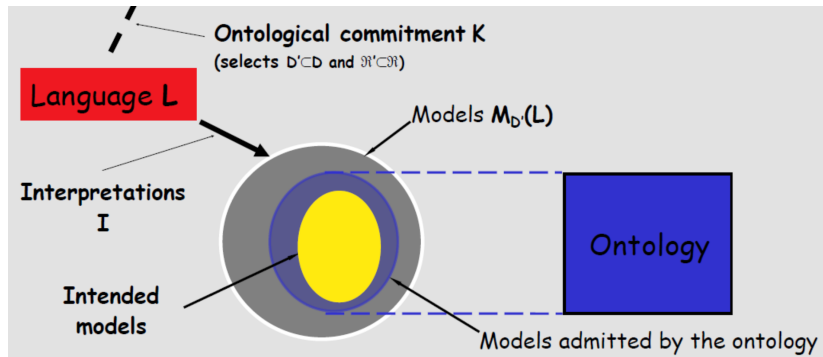


Figure: Simplified version of the definition of ontology based on [Guarino, 2009]

An Ontology is a logical theory such that its formal models approximate the intended interpretation of a vocabulary as good as possible

FOL Decision Problems

Main decision problems:

- **Satisfiability:** φ is satisfiable if it has at least one model
- **Entailment:** $\Gamma \models \varphi$ if every model of Γ is also a model of φ

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The satisfiability problem for FOL is undecidable

- No algorithm can decide truth for all FOL formulas
- Practical impact: automated reasoning may not terminate
- Leads to study of decidable fragments (restricted languages)

Exercises: First-Order Logic

① Formalise the sentences below in FOL

- “All dogs are animals.”
- “Some students read every book.”
- “Every teacher teaches at least one student.”

② Show that $\exists t \left(Teacher(t) \wedge AttendsISAQ(t) \right) \wedge \forall s \forall \ell \left((Teaches(t, s) \wedge Understands(s, \ell)) \rightarrow Understands(t, \ell) \right)$ is satisfiable.

③ Advanced Exercise $\Rightarrow$ Give an example of a formula in FOL that is satisfiable but has no finite model.

Description Logics (DLs)

Student $\sqcap$ *AttendsISAO* $\sqsubseteq \exists$ *understands.Logic*

Description Logics (DLs)

- DLs are fragments of FOL tailored for representing **ontologies**.
- DLs trade expressivity for **decidability and efficient reasoning**.
- Used widely in the **Semantic Web**, e.g., OWL ontologies like **SNOMED CT**.
- They come in different flavours: *ALC*, *SHIQ*, *EL*, DL-Lite, ...
- The different DLs vary in expressivity - complexity

Syntax of DL Concepts

$$C, D ::= \top \mid \perp \mid A \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \forall R.C \mid \exists R.C \mid \dots$$

- **Atomic concepts:** $A, B, C, \dots$ (unary predicates, e.g., Student)
- **Roles:** $R, S, \dots$ (binary relations, e.g., takes, teaches)
- **Individuals:** $a, b, \dots$ (constants, e.g., Mary, Math)
- **Constructors:**
 - $\top, \perp$: universal / empty concepts
 - $\neg C$: negation
 - $C \sqcap D, C \sqcup D$: conjunction / disjunction
 - $\forall R.C$: universal restriction
 - $\exists R.C$: existential restriction
 - $\dots$

Knowledge Bases in DLs

A knowledge base $\mathcal{K}$ consists of:

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- **TBox (Terminological Box):** set of GCIs of the form $C \sqsubseteq D$

Define hierarchies and relationships between concepts

Examples:

- Student $\sqsubseteq$ Person
- Teacher $\sqsubseteq \exists \text{teaches.Course}$

Knowledge Bases in DLs

A knowledge base $\mathcal{K}$ consists of:

$$\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$$

- **TBox (Terminological Box):** set of GCIs of the form $C \sqsubseteq D$
Define hierarchies and relationships between concepts
Examples:
 - $\text{Student} \sqsubseteq \text{Person}$
 - $\text{Teacher} \sqsubseteq \exists \text{teaches.Course}$
- **ABox (Assertional Box):** set of assertions of the form $C(a), R(a, b)$
State facts about specific objects
Examples:
 - $\text{Teacher}(\text{Alice})$
 - $\text{teaches}(\text{Alice}, \text{Math})$

Interpretation

An interpretation $\mathcal{I} = \langle \Delta^{\mathcal{I}}, \cdot^{\mathcal{I}} \rangle$ consists of:

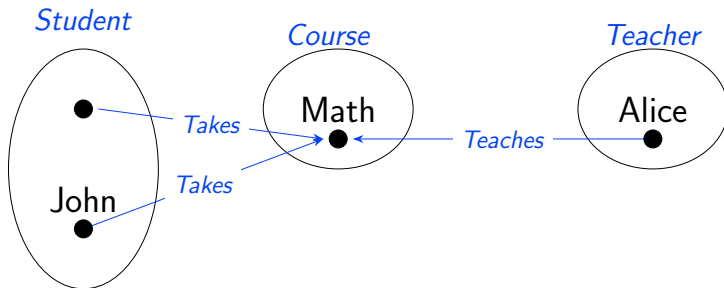
- A non-empty domain $\Delta^{\mathcal{I}}$
- An interpretation function $\cdot^{\mathcal{I}}$ assigning:
 - To each atomic concept A : a subset $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$
 - To each role R : a binary relation $R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$
 - To each individual a : an element $a^{\mathcal{I}} \in \Delta^{\mathcal{I}}$

Semantics of Description Logics

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From DLs back to FOL

- Concepts map to unary predicates: $C \Rightarrow C(x)$
- Roles map to binary predicates: $r \Rightarrow r(x, y)$
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$C \sqsubseteq D$	$\Rightarrow$	$\forall x (C(x) \rightarrow D(x))$
$\neg C$	$\Rightarrow$	$\neg C(x)$
$C \sqcap D$	$\Rightarrow$	$C(x) \wedge D(x)$
$C \sqcup D$	$\Rightarrow$	$C(x) \vee D(x)$
$\exists r.C$	$\Rightarrow$	$\exists y (r(x, y) \wedge C(y))$
$\forall r.C$	$\Rightarrow$	$\forall y (r(x, y) \rightarrow C(y))$

Example:

DL: $\langle \{ Student \sqsubseteq \exists takes.Course \} \{ Course(Math) \} \rangle$

FOL: $\forall x (Student(x) \rightarrow \exists c (takes(x, c) \wedge Course(c))) \wedge Course(Math)$

The Description Logic $\mathcal{EL}$

- $\mathcal{EL}$ is a **lightweight DL**.
- Supports:
 - Conjunctions: $C \sqcap D$
 - Existential restrictions: $\exists R.C$
 - Top concept: $\top$
 - Bottom concept (on the right): $\perp$(No negation, disjunction or universal restrictions)
- Efficient reasoning allows **large-scale ontologies** with millions of entities
- Common in the biomedical domain. Eg: SNOMED CT, Gene Ontology

Reasoning tasks in $\mathcal{EL}$

- **Subsumption:** $C \sqsubseteq D$? Is every C also a D ?
- **Instance checking:** $C(a)$? Is individual a a member of concept C ?
- **Consistency:** Is the ontology free of contradictions?

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In $\mathcal{EL}$, these tasks can be solved in **polynomial time**.

Exercises: Description Logics

- 1 Express in DL:
 - Every professor is a person.
 - Every student attends some course.
 - Courses are taught only by teachers.

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$\text{Student} \sqsubseteq \exists \text{takes.Course}, \quad \text{Teacher} \sqsubseteq \exists \text{teaches.Course}$

$\text{Student}(\text{Bob}), \quad \text{teaches}(\text{Mary}, \text{Math}), \quad \text{Course}(\text{Math})$

Reasoning tasks (entailment):

- Is Mary a Teacher?
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Exercises: Description Logics

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Reasoning tasks (entailment):

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 - Does Bob take Math?
- 3 Construct a DL knowledge base in $\mathcal{EL}$ that describes the animal kingdom (Animals are of disjoint families, Lions eat Animals, etc.).

OWL Ontologies in the Manchester Syntax

The Web Ontology Language (OWL)

- **OWL** is the standard ontology language for the **Semantic Web**.
- Based on **Description Logics** (DLs) $\Rightarrow$ formal semantics and decidable reasoning.
- Comes in different profiles:
 - **OWL 2 DL**: expressive, decidable, based on *SROIQ*
 - **OWL 2 EL** (based on $\mathcal{EL}$), **OWL 2 QL**, **OWL 2 RL**: lightweight profiles for scalability
- Supports:
 - **Classes, Properties, Individuals**
 - Reasoning services: classification, consistency checking, query answering
- Syntaxes: RDF/XML, Turtle, **Manchester Syntax** (human-readable)

- Human-readable syntax for DLs
- The concept constructs of EL:
 - Conjunction: `and (Concept1 Concept2)`
 - Existential restriction: `some Role Concept`
 - Top concept: `Thing`
- GCI constructs:
 - Subsumption: `Concept1 SubClassOf: Concept2`
 - Equivalence: `Concept1 EquivalentTo: Concept2`
 - Disjointness: `Concept1 DisjointWith: Concept2`

TBox axioms:

Class : Concept1

SubClassOf : Concept2

ABox axioms:

Individual : Mary

Types : Student

Facts : takes Math

Example: University Domain

TBox axioms:

Class : Student

SubClassOf : Person

Class : GraduateStudent

*SubClassOf : Student and
(takes some Course)*

Class : Teacher

SubClassOf : Person

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Exercises: Manchester Syntax

① Write in Manchester Syntax:

Mammal $\sqsubseteq$ *Animal*

Bird $\sqsubseteq$ *Animal* $\sqcap$ *Flies*

Animal $\sqsubseteq$ $\exists hasParent. Animal$

Bird $\sqcap$ *Mammal* $\sqsubseteq \perp$

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Class: Bird SubClassOf: Animal and Flies

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Class: Bird DisjointWith: Mammal

2 Extend the “Animal ontology” with at least with these roles: *hasPart*, *hasParent*

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$Mammal \sqsubseteq Animal$

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2 Extend the “Animal ontology” with at least with these roles: *hasPart*, *hasParent*

3 ... finally, an answer to the FOL exercise on forcing infinite models:

$$\exists x \neg \exists y S(y, x) \quad (1)$$

$$\wedge \forall x \exists y S(x, y) \quad (2)$$

$$\wedge \forall x \forall y \forall z ((S(x, z) \wedge S(y, z)) \rightarrow x = y) \quad (3)$$

- Baader et al. *The Description Logic Handbook*
- Hitzler et al. *Foundations of Semantic Web Technologies*
- Horrocks et al. on OWL and DL applications